

Medical Deep Learning Models do Learn About the World

by

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Bachelor Thesis

Computer Science

Submission: May 11, 2026

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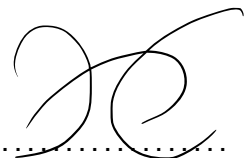
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Preface

The research performed in this thesis was done in collaboration with Bilal Waraich; Sections [3.1](#), [3.2](#) and [3.4](#) (excluding [3.4.5](#)) have been worked on together.

Abstract

Deep learning models in medical imaging are typically treated as black boxes, which is a barrier to clinical adoption: when a diagnosis is wrong, clinicians cannot tell a genuine misclassification from a model latching onto a spurious imaging artefact. Mechanistic interpretability addresses this by reverse-engineering the model's internal computations, but the field has so far focused on language models and natural-image vision transformers. The central question this thesis asks is one of *universality*: does a medical vision model trained without any explicit clinical vocabulary nonetheless develop one?

A Matryoshka sparse autoencoder is applied to UMedPT — a Swin Transformer pre-trained on seventeen medical datasets — and the resulting 4096 features are filtered through a five-stage audit pipeline. Of these, 252 are *mathematically universal*: shared, semantically consistent, and stable across four chest X-ray datasets. Clinical probing then identifies a much smaller *semantically universal* subset of three latents — two healthy-lung detectors and one pleural-abnormality detector — that discriminate above the random baseline on every dataset. UMedPT's pretraining objective does not reward a shared clinical vocabulary, yet an unsupervised dictionary recovers features that behave as if one were present, suggesting that a form of universal medical world knowledge emerges in a model that was never trained to learn one.

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1 Introduction

Mechanistic interpretability (MI), a subfield of explainable artificial intelligence (XAI), is an emerging research field in which researchers attempt to analyse deep neural networks by reverse engineering their internal computations [5]. Unlike post-hoc explanation methods, MI aims to identify a minimal set of components — referred to as *circuits* — that are causally responsible for specific model behaviours [17]. This results in a more transparent and therefore trustworthy model: one that can be audited and reasoned about, rather than treated as an opaque *black box*. The practical significance of this transparency spans many industries in which machine learning models inform high-stakes decisions. In predictive maintenance, for instance, understanding why a model anticipates equipment failure allows engineers to trace and address the underlying failure modes. In medicine, the ability to inspect and verify a model’s reasoning is a prerequisite for clinical adoption [40]: where a model and a clinician disagree on a diagnosis, a confidence score alone is insufficient — an explanation is needed.

Currently, MI research is conducted predominantly on Large Language Models (LLMs), with Anthropic and Google DeepMind being the most active contributors [17, 31]. LLMs are a natural target for this research: pre-trained on large, diverse corpora spanning multiple domains, languages, and disciplines [7], these models encode a rich form of *parametric knowledge* that can be systematically studied. However, for medical imaging tasks, LLMs are not directly applicable — they operate on a fundamentally different input modality and lack the spatial inductive biases required for image analysis. Instead, vision-based architectures such as convolutional neural networks (CNNs) and vision transformers are the standard approach, consistently outperforming language models on medical image benchmarks [40].

MI research on vision-based architectures trained for medical imaging tasks remains largely unexplored [50, 2]. This represents a significant gap: the same tools that have helped researchers understand hallucinations, multi-step reasoning, and refusal behaviour in LLMs have yet to be systematically applied to the medical vision models.

The Swin Transformer [30] is a compelling target for this investigation. Its hierarchical, shifted-window self-attention mechanism shares structural similarities with the multi-head attention studied in LLM circuit research, suggesting that MI techniques may transfer across modalities. In particular, the UMedPT Swin Transformer [42] — a multimodal medical foundation model pre-trained on over seventeen diverse medical datasets spanning different tissues, tasks, and imaging modalities — constitutes a strong candidate: its broad medical pre-training is analogous to the diverse corpora that make LLMs suitable targets for MI, and its transformer architecture provides a direct structural basis for circuit-based analysis.

This thesis investigates whether MI techniques developed for LLMs can be transferred to UMedPT, and whether doing so reveals interpretable circuits encoding clinically relevant medical knowledge. Specifically, we ask: *does UMedPT contain universal features that represent world knowledge in the medical sphere?* The notion of *universality* is borrowed from the circuits literature [36], where it refers to analogous features (A.K.A latents) recurring across networks trained on related tasks. In the present setting, the corresponding question is whether UMedPT’s internal representations have acquired a general concept of pathology that is independent of disease, dataset, and ideally modality.

A *universal* feature is defined as a latent that has the following properties:

Domain invariance. The feature fires consistently and in the same representational direction across datasets: its token-level activation frequency is at least 1% on at least two of the four datasets, and its per-dataset activation centroids in encoder space have pairwise cosine similarity of at least 0.70.

Cross-population robustness. The feature activation is independent of demographic and epidemiological context, i.e. it does not co-vary with patient- or cohort-level covariates that are orthogonal to the pathology.

Spatial grounding. The highest-activating patches of the feature localise with regions that radiologists have verified as containing the pathology, distinguishing pathology-encoding features from those firing on geometric or acquisition artefacts.

Monosemanticity. A single human-interpretable concept is encoded by one feature rather than by a linear combination of features, ruling out polysemanticity and feature splitting as explanations for the observed activity [18, 6].

The remainder of this thesis is structured as follows. Section 2 reviews the relevant literature on ML, transformer architectures, and medical imaging. Section 3 describes the experimental methodology and circuit extraction pipeline. Section 4 presents and evaluates the results. Section 5 summarises the findings and section 6 outlines directions for future work.

2 Literature Review

2.1 Vision Transformer Architectures

2.1.1 From Self-Attention to Patch-Based Vision Models

The transformer architecture, introduced by Vaswani et al. for neural machine translation, operates on self-attention and feed-forward layers, discarding the recurrence and convolution operators that dominated deep learning prior to 2017 [49]. The core operation — scaled dot-product attention — computes a weighted sum of value vectors, where weights are determined by the compatibility between queries and keys. In the multi-head variant, this operation is performed in parallel across multiple learned subspaces, allowing the model to jointly attend to information from different representational perspectives.

Dosovitskiy et al. demonstrated that this architecture could be applied directly to images by partitioning each image into a sequence of fixed-size non-overlapping patches, linearly embedding each patch, and prepending a learnable class token whose final representation is used for classification [16]. This Vision Transformer (ViT) matched or exceeded CNN performance on large-scale benchmarks when trained on sufficiently large datasets. Subsequent work by Touvron et al. showed that competitive performance could be achieved with significantly less data through knowledge distillation from a CNN teacher [48], while Bao et al. adapted BERT-style masked modelling for self-supervised ViT pre-training [3]. Caron et al.'s DINO demonstrated that self-supervised ViT representations spontaneously encode explicit semantic segmentation information, with attention maps closely tracking object boundaries — a property that is directly relevant to interpretability [12].

A notable limitation of plain ViTs is the quadratic complexity of global self-attention with respect to sequence length. Since medical images are often high-resolution volumetric

scans, this computational cost is prohibitive. Darcet et al. further identified artefactual tokens in plain ViT attention maps — positions corresponding to uninformative background regions that accumulate disproportionate attention weight — and proposed register tokens as a fix [15].

2.1.2 The Swin Transformer

The Swin Transformer addresses the computational limitations of global attention by restricting self-attention to *local, non-overlapping windows*, and achieves cross-window communication through a *shifted-window* mechanism that alternates the window partition between consecutive layers [30]. This design yields linear computational complexity in the image resolution and introduces a hierarchical representation, where spatial resolution is progressively reduced while channel dimension grows — analogous to the feature pyramid structure of ResNets [24]. The resulting architecture serves as a strong general-purpose visual backbone, achieving state-of-the-art results on image classification, object detection, and semantic segmentation. Swin Transformer V2 further scales the architecture in both capacity and resolution through normalisation adjustments and a log-spaced continuous position bias [29].

From a mechanistic interpretability perspective, the Swin Transformer’s attention structure introduces both opportunities and complications. The multi-head self-attention within each window is structurally identical to that in standard transformers, and should in principle be amenable to the same circuit-based analysis. However, the windowed attention pattern means that attention heads are restricted in which positions they can attend to, potentially distributing long-range reasoning across multiple layers via the shifted-window pathway rather than across the heads of a single layer. Whether MI techniques originally designed for full-attention models transfer cleanly to this hierarchical, windowed setting is an empirical question that the present work addresses.

2.2 Transformers in Medical Imaging

2.2.1 Architecture Landscape

The U-Net of Ronneberger et al. established the dominant paradigm for medical image segmentation [41]: a contracting encoder path progressively reduces spatial resolution while increasing feature dimensionality, and an expanding decoder path recovers spatial resolution using skip connections from corresponding encoder layers. While highly effective for pixel-level tasks, U-Net and its variants are CNN-based and thus lack the global receptive field that attention provides from the first layer.

Cao et al. replaced both the encoder and decoder of the U-Net with Swin Transformer blocks, demonstrating competitive performance on cardiac and abdominal CT segmentation benchmarks [11]. Hatamizadeh et al. combined a Swin Transformer encoder with a convolutional decoder, achieving top results on the BraTS 2021 brain tumour segmentation challenge [23]. Tang et al. developed self-supervised pre-training strategies for 3D Swin Transformers on unlabelled medical volumes, showing that pre-training on in-domain data substantially improves downstream segmentation performance [46]. Shamshad et al. provide a comprehensive survey of transformer applications across the full spectrum of medical imaging modalities [43].

2.2.2 UMedPT: A Foundation Model for Medical Imaging

The model that constitutes the primary object of study in this thesis is UMedPT (Universal Medical Pre-trained Transformer), introduced by Schäfer et al. [42]. UMedPT is a Swin Transformer pre-trained jointly on seventeen diverse medical imaging datasets spanning different tissues, imaging modalities (MRI, CT, X-ray, ultrasound), and annotation types. This multi-task pre-training strategy is designed to produce general-purpose visual features that transfer efficiently to downstream clinical tasks.

The breadth of UMedPT’s pre-training is central to its suitability as a subject for MI analysis. A model trained narrowly on a single task or modality may encode relatively task-specific representations that are difficult to generalise from. UMedPT’s exposure to diverse medical domains creates a richer internal representation space — one that may encode general-purpose medical visual concepts in a manner more analogous to how LLMs encode general semantic knowledge. This diversity is a prerequisite for the kind of knowledge circuit analysis that this thesis attempts: for circuits to be meaningful, the model must represent a sufficiently broad range of concepts that the circuits can be compared and distinguished.

2.2.3 Clinical Need for Interpretability in Medical Imaging

Despite the strong empirical performance of transformer-based models on medical imaging benchmarks, their clinical deployment remains constrained by the lack of transparency. Patrício et al. survey a broad range of explainability approaches applied to deep learning in medical image classification, highlighting the gap between post-hoc explanation methods — gradient-based saliency maps, class activation mappings, and attention visualisations — and the mechanistic understanding that would be required for clinical trust [40]. Wang et al. make the specific argument that MI is nascent in the medical imaging domain, and that the field lacks the systematic circuit-level analyses that have been conducted for language models [50].

A central limitation of existing XAI methods in medical imaging is that they explain predictions in terms of input features (which pixels influenced the decision) rather than in terms of internal computations (what the model learned to detect). Saliency maps, for instance, can indicate that a model attended to a lesion boundary, but cannot explain *why* the model treated that boundary as diagnostically relevant or whether the underlying representation is faithful to medical knowledge or to a spurious correlation in the training data. Evaluations of explainability methods on medical ViTs confirm that different methods produce substantially different explanations for the same model [4], underscoring the need for ground-truth mechanistic analyses rather than attribution heuristics.

2.3 Mechanistic Interpretability

2.3.1 The Circuits Paradigm

Mechanistic interpretability (MI) is a research programme that aims to reverse-engineer trained neural networks into human-understandable “features” [32, 5]. Rather than treating a model as a black box and probing its input–output behaviour post hoc, MI seeks to identify the internal structures — *features* and *circuits* — that causally mediate specific model behaviours. A *feature* is a direction in activation space that corresponds to a coherent, human-interpretable concept; a *circuit* is a subgraph of the computational graph

along which features interact to produce a particular output [37].

The circuits programme originated in the context of convolutional neural networks (CNNs). Olah et al. demonstrated that individual neurons in InceptionV1 encode interpretable visual concepts such as curve detectors, high-frequency detectors, and multimodal neurons [33, 35, 37]. Crucially, these features were shown to be *universal*: the same curve-detector circuits appear independently across diverse CNN architectures trained on different datasets [37, 10], suggesting that the representations learned by gradient descent reflect structural regularities of the data rather than idiosyncrasies of architecture or training.

This universality hypothesis motivated the extension of the circuits framework to transformer-based architectures. Elhage et al. introduced a mathematical framework that decomposes transformer computations into a series of residual stream updates, enabling precise attribution of model behaviour to individual attention heads and MLP layers [17]. Within this framework, *induction heads* — pairs of attention heads that implement a pattern-matching algorithm for in-context learning — were identified as a concrete example of how circuits generalise across transformer scales [17]. Subsequent work extended this methodology to more complex behaviours, most notably indirect object identification in GPT-2 small, where a circuit of 26 attention heads was manually traced and causally validated using activation patching [5]. The field has since developed automated circuit discovery methods, most prominently ACDC (Automated Circuit Discovery), which iteratively prunes edges from the computational graph to isolate the minimal subgraph responsible for a target behaviour [13].

2.3.2 Superposition and Polysemanticity

A foundational challenge for ML is that neural networks routinely represent far more concepts than they have neurons. This phenomenon, termed *superposition*, arises because networks must encode a large number of approximately sparse features using a smaller number of dimensions, and can do so without significant loss provided the features are sufficiently infrequent [18]. As a consequence of superposition, individual neurons tend to respond to multiple unrelated concepts — a property known as *polysemanticity* [18]. Polysemantic neurons resist straightforward interpretation and make circuit extraction substantially harder, since individual activations can no longer serve as reliable proxies for specific features.

Complementary theoretical work has examined the geometry of the resulting representations. The *linear representation hypothesis* posits that high-level semantic concepts are encoded as linear directions in activation space [39], a claim supported by the success of linear probing and model steering across a wide range of models. However, this view has been partially challenged by Engels et al., who demonstrate that some features — particularly those encoding cyclically structured concepts — occupy multi-dimensional, non-linear subspaces [19]. Characterising which features are linear and which are not remains an active area of investigation.

2.3.3 Sparse Autoencoders

The dominant approach to resolving superposition is *dictionary learning* via sparse autoencoders (SAEs). An SAE is a two-layer network trained to reconstruct model activations from a sparse, overcomplete latent code: it consists of an encoder that projects

activations into a high-dimensional space and applies a sparsity-inducing activation function, and a decoder whose columns form a learned dictionary of features [6]. Formally, for an input $x \in \mathbb{R}^n$ the encoder and decoder are defined as [9]:

$$f(x) = \sigma(\mathbf{W}^{enc}\mathbf{x} + b^{enc}) \quad (1)$$

$$\hat{\mathbf{x}} = \mathbf{W}^{dec}f(\mathbf{x}) + b^{dec} \quad (2)$$

The SAE’s general loss function is defined as:

$$\mathcal{L}(\mathbf{x}) = \underbrace{\|\mathbf{x} - \hat{\mathbf{x}}(f(\mathbf{x}))\|_2^2}_{reconstruct} + \underbrace{\lambda S(f(\mathbf{x}))}_{sparsity} + \alpha \mathcal{L}_{aux} \quad (3)$$

where λ and α are manually tuneable hyperparameters and $\mathcal{L}_{aux}$ is the standard auxiliary loss used in Gao et al. [20].

The key insight is that, when trained with an appropriate sparsity penalty, the latent dimensions of the SAE correspond to the features that the model itself represents in superposition — each latent dimension is active only for the specific contexts in which the corresponding feature is present [14]. The resulting decomposition is substantially more monosemantic than the raw neuron basis.

Bricken et al. demonstrated this principle on a one-layer transformer, extracting tens of thousands of interpretable features including tokens, named entities, and syntactic patterns [6]. Templeton et al. subsequently scaled this methodology to Claude 3 Sonnet, a frontier language model, recovering features corresponding to abstract concepts such as specific people, cities, and moral sentiments [47]. These results established SAEs as the standard tool for large-scale feature extraction in ML.

Subsequent work has refined both the architecture and training of SAEs. The original formulation uses an L_1 penalty on latent activations to encourage sparsity, but this induces *activation shrinkage*: the magnitude of latent activations is systematically underestimated relative to the optimal reconstruction. Currently there are several different SAE architectures with the most recent ones being BatchTopK, Gated and Matryoshka. This work focuses on the Matryoshka architecture and is explored in more depth in Section 3.3.¹ The broader survey by Shu et al. provides a comprehensive treatment of SAE architecture variants, training strategies, evaluation metrics, and downstream applications [44].

2.3.4 Activation Patching

A core experimental technique in ML is *activation patching* (also called *causal tracing* or *interchange interventions*). In an activation patching experiment, the activations of a target model run are replaced at a specific layer and position with activations from a *corrupted* or *counterfactual* run, and the effect on the model output is measured [52]. This allows researchers to identify which components — attention heads, MLP layers, or individual residual stream positions — are causally necessary for a given behaviour. Zhang et al. provide a systematic analysis of the methodological choices involved in activation patching, including the choice of corruption method (zero ablation, mean ablation, or noise patching) and the choice of output metric, demonstrating that these choices materially affect conclusions [52]. Heimersheim offers a practical guide to applying these techniques in practice [25].

¹A parallel thesis by Bilal Waraich investigates universality within UMedPT using the BatchTopK architecture instead.

2.4 Mechanistic Interpretability for Vision Models

2.4.1 CNN-Era Circuit Analysis

The foundations of MI for vision were established in the context of CNNs. Olah et al.’s Distill series demonstrated that the internal representations of InceptionV1 decompose into interpretable features organised into circuits [33, 35, 37, 34, 10]. These analyses established the methodology of *feature visualisation* — generating synthetic inputs that maximally activate a given neuron using activation maximisation — and *circuit dissection* — tracing information flow through the network to understand how high-level features are composed from lower-level ones. While powerful, these methods were limited by the polysemanticity of individual neurons: many neurons in CNNs respond to multiple unrelated visual concepts, making clean circuit extraction difficult.

2.4.2 ViT Interpretability

Extending MI to vision transformers raises distinctive challenges. Unlike CNNs, where spatial processing is local by construction, ViTs compute global self-attention across all patch positions from the first layer. The resulting attention patterns are often difficult to interpret directly: raw attention weights do not constitute a faithful explanation of information flow, as they ignore the magnitude of value vectors and the composition of information across layers. Bahador examines the attention heads of fine-tuned ViTs on distorted images and finds monosemantic behaviour in intermediate layers alongside evidence of functional specialisation, including heads that detect edges, textures, and high-level object parts [2]. This suggests that the monosemanticity and functional specialisation observed in LLM circuits may have analogues in vision transformers.

Nanda et al.’s analysis of grokking via MI provides a further example of how circuit-based analysis can reveal qualitatively new understanding of model behaviour: by tracking how circuits form over training, they demonstrated that grokking corresponds to the emergence of a specific algorithmic circuit, replacing memorisation with generalisation [31]. Progress measures of this kind may be applicable to understanding how medical knowledge consolidates in ViTs over pre-training.

2.4.3 Sparse Autoencoders for Vision Models

The application of SAEs to vision models is an emerging and rapidly developing area. Interpretable and Testable Vision Features via Sparse Autoencoders demonstrates that a single SAE trained on a frozen ViT’s residual stream supports patch-level causal edits across both classification and segmentation tasks, establishing that SAE features in vision models are causally meaningful rather than merely correlational [45]. Pach et al. extend the monosemanticity result to vision-language models, showing that SAE-based feature decomposition recovers interpretable visual concepts including object parts, textures, and semantic categories [38].

Joseph et al. apply SAE analysis to CLIP’s vision transformer, uncovering fundamental differences between the mechanisms by which vision and language transformers process information — a finding with direct implications for the transferability of LLM MI methods to vision models [27]. Critically, a causal analysis of SAE features in ViTs by Han, Kim, and Kwak finds that a substantial proportion of features cannot be understood from their activation patterns alone: the causal effect of a feature on model outputs often differs

from what its maximum-activating inputs would suggest [21]. This motivates the use of causal interventions — rather than inspection of activation statistics — as the primary method of feature validation.

2.4.4 The Research Gap

Despite this recent progress, the mechanistic interpretability of vision transformers trained on medical imaging tasks remains unexplored. Existing ViT MI work has focused on models trained on natural images (ImageNet, CIFAR, or CLIP’s web-scale dataset); the representations learned by these models differ substantially from those of models trained on structured, domain-specific medical images. In particular, the features relevant to medical imaging — tissue textures, anatomical boundaries, pathological patterns, and modality-specific artefacts — have no natural analogues in the general-domain literature, and there is no *a priori* reason to expect that the SAE dictionaries or circuit structures discovered for natural-image models will transfer.

Furthermore, no prior work has examined whether knowledge circuits of the kind identified in LLMs by Yao et al. [51] emerge in vision transformers at all, let alone in medical ones. The present thesis makes a first attempt to fill this gap, applying state-of-the-art SAE-based feature extraction and circuit analysis to UMedPT in order to determine whether its internal representations organise into interpretable, causally meaningful circuits encoding clinically-relevant medical knowledge.

3 Methodologies

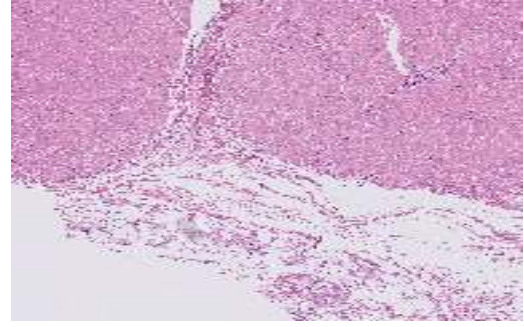
3.1 Datasets and modality

UMedPT is pretrained on over seventeen datasets spanning multiple organs, lesion types, and imaging modalities [42], allowing flexibility in dataset selection. Three criteria governed the choice: (1) *volume* — sufficient samples to populate a diverse activation corpus; (2) *concept diversity* — variation in pathology or anatomy to encourage a rich set of learned features; and (3) *interpretability* — a modality whose image content is legible to a non-specialist evaluator, enabling human validation of the features extracted by the SAE.

Histopathological slides, for example, satisfy criteria (1) and (2) but fail criterion (3): the sub-cellular structures that distinguish tissue classes are not readily interpretable without domain expertise. Plain radiographs, by contrast, depict gross anatomical structures — lung fields, the cardiac silhouette, bony landmarks — that are identifiable without specialist training (Figure 1). MRI was also considered but excluded on two grounds: the UMedPT corpus contains substantially fewer MRI samples than radiographs, and the MRI acquisitions cover heterogeneous organs and protocols, reducing the availability of pathology-matched ground-truth labels useful for downstream probing experiments. Furthermore, the MRI modality splits a 3D object into multiple sequential 2D slices, in contrast to X-ray imaging which is a simple 2D image, making the latter easier to interpret because a single image contains all the information needed. Chest X-rays were therefore selected as the primary modality: the dataset is large, the pathology labels (e.g. bacterial vs. viral pneumonia, normal) provide clear ground truths for concept verification, and the images contain salient anatomical structure at a spatial resolution amenable to patch-level analysis.



(a) Chest X-ray, bacterial pneumonia



(b) Colon histopathology slide.

Figure 1: Comparison of two candidate modalities.

Four chest X-ray datasets were chosen: CheXpert — over 200,000 posteroanterior (PA) images from the Stanford Hospital which include fourteen different labels (e.g. “lung lesion”, “pneumonia”, “fracture”) [26]; Shenzhen TB — 600 images from the Shenzhen No. 3 People’s Hospital in China with a binary classification of “tuberculosis” and “normal”; SIIM-COVID-19 — containing over 6,000 unique COVID-19 studies where each image represents one study and has four labels (*negative*, *typical*, *indeterminate*, *atypical*); and RSNA — a dataset with over 26,000 PA images with a binary classification of “pneumonia” and “normal”. These labels will be used in Section 4 to see if features activate to specific diseases, thus making the feature analysis more concrete as we can analyse where on the image a feature is being activated and for what type of illnesses or none.

The X-ray images come in different resolutions and are typically grey-scaled; however, due to the nature of UMedPT and the Swin Transformer, the expected images have a resolution of 224×224 pixels and three channels (RGB) instead of one. Each image used during this research has been modified to meet these expectations.

It is evident that there is a problem with the distribution of images. The large volume of images coming from CheXpert will cause any SAE trained on these four datasets to have a bias towards CheXpert, as the activations coming from said dataset are going to be much greater in volume than those from smaller sets such as Shenzhen TB. Figure 2 displays the gap in size between each dataset.

If an SAE were to be trained on this data as is, there would be, for example, significantly fewer features and activations recovered from the Shenzhen TB dataset. To ensure an equal distribution of images shown to UMedPT when extracting activations and training the SAE, stratification and subsampling must occur. Stratification is the process of sampling *with replacement* from the smaller datasets so that they can appear in each batch: per epoch, duplicate images, for example, from Shenzhen TB may appear across different batches, while CheXpert images are more likely to be shown only once across all batches in the epoch. Downsampling places a quota on the number of images used during harvesting and training; in this thesis a quota of 10,000 was chosen as it provides sufficient representation without becoming computationally prohibitive. This ensures that each dataset contributes a quarter of all activations and training examples, balancing the bias introduced by source-specific volume.

Figure 2: Visualisation of X-ray datasets sizes (rounded).

3.2 Activation Harvesting

3.2.1 Activation Extraction

As described in Section 2.2, the UMedPT backbone is a Swin Transformer organised into four hierarchical stages, with each stage boundary implemented by a *PatchMerging* layer that halves the spatial resolution and doubles the channel dimension. The choice of where to tap the network determines the semantic character of the resulting activations and, by extension, the features the SAE can recover.

Activations are extracted using a *forward hook* — a callback registered on a specific module that fires during the forward pass as execution reaches that layer, exposing the layer’s output to an external observer. In this work, the hook copies the output tensor into an external buffer on each forward pass, leaving the encoder’s parameters and computation graph unmodified. This non-invasive extraction is preferable to alternatives such as rewriting the model to return intermediates or re-implementing a truncated forward pass, both of which risk silently diverging from the original architecture and complicate the loading of pretrained weights. Once harvesting is complete, the hook is deregistered and the encoder reverts to its original behaviour.

The hook is attached to the *PatchMergingV2* module at the Stage 3→4 boundary, yielding activations of dimension $d_{\text{model}} = 1024$ at a spatial resolution of 7×7 tokens per image. Thus, the image is subdivided into 49 regions of 32×32 pixels, illustrated in Figure 3.

Two considerations motivate this placement. First, it captures the deepest spatially-resolved representation in the network: any later tap would follow the global pooling operation that collapses the token grid into a single embedding, precluding the patch-level feature visualisation central to this work. Second, representations at this depth are sufficiently abstract to encode pathology-level concepts rather than low-level texture, which is

Figure 3: The 7×7 patch grid segmentation of an image.

the regime in which SAEs have been shown to recover human-interpretable features [9].

3.2.2 Spatial Flattening

The tensor captured by the forward hook has shape $(B, 7, 7, d_{\text{model}})$: for each of the B images in the batch, the Swin encoder produces a 7×7 grid of 1024-dimensional tokens, each token corresponding to one non-overlapping region of the original image. The SAE, by contrast, is a fixed-width two-layer transformation defined on single vectors; it has no mechanism for ingesting a spatial grid. The hook output must therefore be reshaped into a flat sequence of token vectors before training:

$$\text{reshape} : (B, 7, 7, d_{\text{model}}) \mapsto (49B, d_{\text{model}}) \quad (4)$$

The operation is purely an index rearrangement — the spatial correspondence between a flattened token and its source position is preserved and can be recovered exactly by inverting the reshape. Each of the $49B$ resulting vectors is treated as an independent training example for the SAE: the autoencoder is not told which tokens share a source image, nor where in the grid a token originated.

Three considerations motivate this choice. First, it is the minimal modification required by the SAE architecture, and it discards no information — the spatial axes are moved rather than collapsed. Second, training on position-agnostic tokens is the correct inductive bias for the features this work seeks to recover. A disease concept such as pleural effusion or airspace opacity manifests in different spatial locations across patients and acquisition protocols; by withholding positional information during SAE training, the dictionary is encouraged to learn features that respond to the *content* of a patch rather than to its coordinates. Third, the flattening multiplies the effective training volume by a factor of 49: from the roughly 26,700 images across the four datasets, the SAE is trained on approximately 1.3×10^6 token vectors. This amplification is necessary because sparse dictionaries with thousands of latents would suffer from dead latents when trained on insufficient tokens [20].

Although spatial information is withheld at training time, it is recoverable at analysis time. For any image and any learned feature, the 49 activations of that feature on that image’s tokens can be reshaped back into a 7×7 map, yielding a coarse spatial heatmap of where the feature fired. This map can be upsampled to the input resolution and compared against radiologist bounding boxes (where available, as in SIIM-COVID-19 and RSNA) to

verify that features identified as encoding a given pathology fire at anatomically plausible locations.

3.2.3 Activation Normalisation

This thesis harvests the activations of each dataset separately and then concatenates them together. However, before the concatenation can occur, the activation vectors must be normalised. Each hospital varies in the equipment used, the position of the equipment and the patients themselves. As a result, two identical lungs imaged at different hospitals with different exposure settings will yield activation vectors pointing in the same direction but at very different scales. Gao et al. proposed to fix this using RMS normalisation [20], given below:

$$x = \frac{x}{\text{RMS}(x) + \varepsilon} \cdot \sqrt{d_{\text{model}}} \quad \varepsilon > 0 \quad (5)$$

$$\text{RMS}(x) = \sqrt{\text{mean}(x^2)} \quad (6)$$

The RMS normalisation is applied at the moment each activation vector is emitted from UMedPT and is agnostic to the dataset from which it originated.

A separate issue lies in the centring of activations: for the reasons above, each dataset can exhibit a different offset in activation space. To address this, a per-dataset mean is subtracted:

$$x_{\text{centered}} = x - \text{mean}(x) \quad (7)$$

where the mean is computed independently over the activations of each dataset. With the activations harvested, the next step is to train the SAE.

3.3 The Matryoshka SAE

As mentioned before, SAE and dictionary based learning has so far shown the most fruition in the field of mechanistic interpretability. However, choosing the dictionary size of the SAE architecture in use has become a point of tension [9]. Unlike traditional deep learning models, scaling does not always improve performance. On one hand, increasing the scale of an SAE does lead to lower training loss; however, its downstream task performance — model behaviour probing, feature circuit analysis, and more — suffers. Before the Matryoshka SAE, the recommended approach was to train a variety of SAEs with different dictionary sizes, which is costly [28].

The Matryoshka sparse autoencoder addresses the dictionary size issue by decoupling the deconstruction and reconstruction into multiple nested SAEs increasing in size. Each SAE is trained to reconstruct the input using only a subset of the latents up to that nested layer. This hierarchical approach prevents the later-scale latents from learning the broad concepts that the smaller-scale latents should encode.

The SAE architecture uses BatchTopK activations which originate from the BatchTopK SAE. The BatchTopK activation function considers the sparsity across the batch rather than individual examples. Given a batch $X = \{x_1, \dots, x_B\}$ — where B is the batch size and K is the per-sample sparsity target — BatchTopK retains the top $B \times K$ largest activations across the entire batch while setting the others to zero, as seen in Equation 8:

$$\text{BatchTopK}(X) = X \odot \mathbf{1}[X \geq \tau(X)] \quad (8)$$

where $\odot$ denotes element-wise multiplication and $\tau(X)$ is the threshold given by the $(B \times K)$ -th largest value in X [8].

The deconstruction function of the matryoshka SAE is the same as the standard for most, however, the reconstruction function is what makes this SAE differ. Given the dictionary size k ² a nested sequence $M = m_1, m_2, \dots, m_n$ where $m_1 < m_2 < \dots < m_n = k$. Each m_i represents a nested SAE that must reconstruct the input using the first m_i features as seen in Equation 9:

$$\mathcal{L}(\mathbf{x}) = \sum_{m \in \mathcal{M}} \underbrace{\left\| \mathbf{x} - \left(\mathbf{f}(\mathbf{x})_{0:m} \mathbf{W}_{0:m}^{\text{dec}} + \mathbf{b}^{\text{dec}} \right) \right\|_2^2}_{\text{reconstruction using first } m \text{ latents}} + \alpha \mathcal{L}_{\text{aux}} \quad (9)$$

Figure 4: Overview of the Matryoshka Sparse Autoencoder pipeline.

As a result of the sub-SAE reconstruction up to the i th feature, a regularisation effect is created, enforcing the Matryoshka SAE “to learn features at multiple levels of abstraction” [9].

3.3.1 Training Configuration

A single Matryoshka SAE is trained on the conglomerate of activations, with batches drawn round-robin from the per-dataset buffers so that every optimisation step sees an

² k can be either chosen as a tuneable hyper-parameter or chosen randomly from a sampled distribution per batch [9]. For this thesis k was manually chosen to be 48 through trial and error.

equal share of each source. The dictionary width is set to $d_{\text{sae}} = 4 \cdot d_{\text{model}} = 4096$ — a four-times expansion of the 1024-dimensional encoder activations, sized to give superposition enough room to be unfolded without making the audit pipeline computationally prohibitive. The Matryoshka prefix schedule is $M = \{6, 12, 24, 48\}$, octave-spaced from a coarse six-feature reconstruction to the 48-feature largest scale at which the BatchTopK activation operates; auxiliary-loss weights α_i are set to $1/k_i$ and renormalised so that the four scale terms contribute equally to $\mathcal{L}_{\text{aux}}$.

Hyperparameter	Value
Encoder dimension d_{model}	1024
Dictionary size d_{sae}	4096 (expansion factor 4)
Prefix schedule M	$\{6, 12, 24, 48\}$
Activation function	BatchTopK ($k = 48$)
Auxiliary-loss coefficient α	$1/32$
ℓ_1 coefficient λ	2×10^{-4}
Dead-latent window	10 batches
Dataset sampling	Round-robin across the four X-ray sets

Table 1: Hyperparameter configuration of the Matryoshka SAE.

3.4 Universality Analysis

This subsection describes the hierarchical pipeline used to identify universal features within the trained Matryoshka SAE. Each stage is gated on the result of the preceding one, with the cheaper checks running first so that the more expensive audits are restricted to a progressively smaller candidate set.

3.4.1 Scanner-Feature Audit

The first step is to identify and prune out features which are trained to focus on scanners and physical equipment rather than more general features. We hypothesise that such latents exist that search for properties of scanners, image resolution, etc. To verify this, a logistic regression is fit for each latent and measures how well each individual latent can predict the dataset of the image. The output of logistic regression is an area under the receiver operating characteristic curve (AUROC) with potential values in $[0, 1]$. If the AUROC value of a latent exceeds 0.8, it is likely that the latent has memorised properties of the imaging technology and not concepts of illnesses, lesions, or anything that is universal. Therefore all features that have an AUROC greater than 0.8 are excluded at this stage of the evaluation.

3.4.2 Feature Activation Frequency

The next step of the process in order to identify if a feature is universal is to look at its activations. The activation rate is computed per dataset, expressed as a percentage, and calculated as follows:

$$\text{frequency}(\text{feature}_i, \text{dataset}_d) = \frac{\#(\text{tokens in dataset}_d \text{ where feature}_i > 0)}{\#(\text{tokens in dataset}_d)} \quad (10)$$

A candidate universal feature is required to activate $\geq 1\%$ per dataset for two or more datasets. This lower bound ensures that the feature is active often enough to carry a real signal rather than noise — a feature firing on only 0.01% of tokens, for instance, is indistinguishable from sporadic noise — while still being selective enough to avoid trivially broad detectors such as edge or background patches. By requiring the feature to be activated on multiple datasets, it ensures that features specialised on singular datasets are filtered out at this stage, allowing only features that detect concepts across modalities or illnesses to proceed.

3.4.3 Cross-Dataset Semantic Consistency

After a feature passes the activation-frequency filter introduced above, it is known to fire with non-negligible density on at least two datasets; however, frequency alone does not guarantee that the concept encoded is the same across those datasets. A latent could, for instance, fire on unremarkable lung texture in one dataset and on radiographic gridding in another — two distinct concepts that nonetheless produce comparable activation rates. The purpose of this stage is therefore to verify that the feature’s *semantic content*, as represented in UMedPT’s activation space, is consistent across the datasets on which it fires. This process also ensures that the features from the SAE are not polysemantic.

For each candidate feature f and each dataset d , let $\mathcal{P}_f^{(d)}$ denote the set of the top- T patches within dataset d on which feature f attains its highest activation (with $T = 50$ in this work), and let $\mathbf{x}_p \in \mathbb{R}^{d_{\text{model}}}$ denote the UMedPT activation vector of patch p at the Stage 3→4 boundary defined in Section 3.2. The *per-dataset centroid* of feature f on dataset d is defined as the mean activation across its top- T patches:

$$\mathbf{c}_f^{(d)} = \frac{1}{|\mathcal{P}_f^{(d)}|} \sum_{p \in \mathcal{P}_f^{(d)}} \mathbf{x}_p. \quad (11)$$

Let $\mathcal{D}_f$ denote the set of datasets on which feature f passed the preceding frequency filter. Cross-dataset semantic consistency is then measured as the minimum pairwise cosine similarity between the per-dataset centroids:

$$\text{sim}(f) = \min_{\substack{d_i, d_j \in \mathcal{D}_f \\ d_i \neq d_j}} \frac{\langle \mathbf{c}_f^{(d_i)}, \mathbf{c}_f^{(d_j)} \rangle}{\|\mathbf{c}_f^{(d_i)}\| \|\mathbf{c}_f^{(d_j)}\|}. \quad (12)$$

A feature is retained as a candidate universal feature when $\text{sim}(f) \geq \tau_{\text{sem}} = 0.70$, a threshold that mirrors the pairwise-centroid criterion of the universal-feature definition given in the introduction. The minimum — rather than the mean — over dataset pairs enforces a strict consistency requirement: a feature that is coherent between two datasets but diverges on a third is rejected, as divergence on any pair falsifies the claim that the feature encodes a single concept across the corpus on which it fires. Features that survive both the frequency and semantic-consistency filters form the candidate universal set submitted to the subsequent split-feature audit.

3.4.4 Split Feature Audit

Even though Matryoshka should prevent feature splitting [9], the SAE could have caused feature splitting across the datasets. Therefore, co-activation tests such as the Jaccard

similarity must take place to identify split features. When a pair of features (f_a, f_b) is flagged as co-activating at the largest scale k_n of the schedule, the nested structure of the Matryoshka SAE permits a further resolution step: tracing the split back to a single *ancestor* feature at a smaller scale $k' < k_n$, if one exists. Let $\mathcal{T}_f^{(k)} \subseteq \{1, \dots, N\}$ denote the indices of the T images on which feature f attains its highest per-image activation when the encoder is restricted to its top- k latents (with $T = 50$ in this work), and let the Jaccard similarity between two such sets be

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (13)$$

A pair (f_a, f_b) is declared *split* at scale k_n when

$$J(\mathcal{T}_{f_a}^{(k_n)}, \mathcal{T}_{f_b}^{(k_n)}) \geq 0.5 \quad (14)$$

For each such split, the remaining scales $k_1 < k_2 < \dots < k_{n-1}$ are examined in ascending order, and a feature g is declared an ancestor of the pair at scale k' when it meets both descendants' top-image sets:

$$J(\mathcal{T}_g^{(k')}, \mathcal{T}_{f_a}^{(k_n)}) \geq 0.4 \quad \wedge \quad J(\mathcal{T}_g^{(k')}, \mathcal{T}_{f_b}^{(k_n)}) \geq 0.4 \quad (15)$$

The smallest scale k^* at which such a g is found yields the ancestor representative, and the descendants f_a, f_b are replaced by g in the universal-feature count, providing direct evidence that the nested hierarchy has recovered a concept shared across scales. Where no ancestor satisfies Equation 15 at any $k' < k_n$, the pair is merged into a single cluster whose representative is the member with the highest mean activation frequency, following the scale-agnostic fallback of the split-feature audit. As a result, all features which are classified as “split” will be identified such that, if any of the split features are identified to be universal, there will be no duplicate counting of universal features within UMedPT.

3.4.5 Prefix-Stability Audit

The split-feature audit removes redundancy across distinct latents at the largest schedule scale, but it does not test whether a single latent encodes a stable concept across the nested prefixes of the Matryoshka SAE itself. A latent that is recovered only at the largest scale k_n but has no coherent counterpart at smaller scales is brittle: it exists because the dictionary budget was large enough to afford it, not because the underlying concept is robustly selected by the encoder at every level of granularity. Since universality is intended to capture features that the model represents *independently* of dictionary size, such latents are excluded.

For each candidate f surviving the split-feature audit, the Jaccard similarity between the top- T image sets at every adjacent pair of scales is computed:

$$J_f^{(k_i, k_{i+1})} = J(\mathcal{T}_f^{(k_i)}, \mathcal{T}_f^{(k_{i+1})}), \quad k_i, k_{i+1} \in \{6, 12, 24, 48\}, \quad (16)$$

where $\mathcal{T}_f^{(k)}$ and the Jaccard coefficient J are defined as in Equation 13. A latent is declared *prefix-stable* when every adjacent-scale Jaccard exceeds the stability threshold:

$$\min_i J_f^{(k_i, k_{i+1})} \geq 0.40. \quad (17)$$

The minimum across adjacent pairs — rather than the mean — enforces stability at every scale transition, so a latent whose top-image set drifts between any two adjacent prefixes is rejected even if the remaining transitions are tight. Latents that survive the prefix-stability audit form the final confirmed universal set submitted to clinical probing.

3.4.6 Illness Stratification

The confirmed universal set of Section 3.4 establishes *that* a latent is shared across datasets but not *what* it encodes. To assign a clinical interpretation, every image is labelled with one of seven canonical illness groups derived from the Fleischner Society glossary [22] — a shared vocabulary onto which the heterogeneous native labels of CheXpert, RSNA, SIIM-COVID-19 and Shenzhen TB are mapped — and the SAE encoder is applied patch-wise to the harvested UMedPT activations, restricted to the K confirmed latents, and reduced to one activation per image by spatial max-pooling:

$$z_{n,k}^{d,\text{img}} = \max_{p \in \{1, \dots, P\}} \tilde{Z}_{\text{patches}}^d[n, p, k]. \quad (18)$$

Max-pooling rather than averaging is the appropriate aggregation because pathological findings occupy only a handful of the $P = 49$ patches; mean-pooling would dilute the signal across the non-pathological remainder.

Stratification then operates on two axes. First, activation statistics are sliced by illness group: for each canonical group g , dataset d , and confirmed latent k , the per-group mean $\mu_{g,k}^d$ and frequency $\phi_{g,k}^d$ are computed independently, yielding a $7 \times K$ illness–latent matrix that summarises how strongly each confirmed latent responds to each clinical concept. Second, a one-vs-rest logistic regression is fit per latent on the scalar feature $x_n = z_{n,k}^{d,\text{img}}$ with class-balanced weighting, and its generalisation is estimated by 5-fold *stratified* cross-validation: fold assignment preserves the per-group proportions, which is necessary because the smallest dataset contains ~ 700 images and a non-stratified split can leave a rare group entirely absent from training. The reported score is the macro one-vs-rest AUROC averaged over valid folds, with a random baseline of 0.5. Linear — rather than nonlinear — probing is used by design [1]: a high score under a linear classifier licenses the conclusion that the latent’s *magnitude* is itself the clinically informative quantity, not merely some monotone function of it.

3.4.7 Manual Inspection

Passing every statistical filter does not by itself certify a latent as universal: a feature may attain the requisite frequency, centroid alignment, and discriminative AUROC while still firing on anatomically distinct regions across datasets, or on semantically vacuous patches such as image borders or lettering. A final manual inspection therefore screens the surviving candidates against two qualitative criteria. (1) *Cross-dataset semantic consistency*: the feature’s top-activating patches must depict the same anatomical or pathological content across datasets; a latent that fires on rib structures in one dataset and on background regions in another is rejected as polysemantic. (2) *Semantic meaningfulness*: the activating content must correspond to a clinically interpretable structure — ribs, lung parenchyma, lesions, pleural margins — rather than to acquisition artefacts that the statistical filters cannot detect.

The inspection is supported by a patch-ablation heatmap, computed by zeroing each of the $P = 49$ tokens in turn and recording the change in the feature’s activation. Over-

laying this heatmap on the original image localises the input region most responsible for the feature’s firing, and is used both to confirm that the activating content is anatomically plausible and to discard candidates whose ablation maps concentrate on borders, corners, or annotation artefacts.

4 Evaluation of the Investigation

4.1 Audit Pipeline Results

The hierarchical pipeline of Section 3.4 reduces the 4096-latent dictionary to a small set of confirmed universal features. This subsection reports the count surviving each stage and the per-dataset statistics that justify the thresholds chosen in the methodology.

4.1.1 Scanner-Feature Audit

Of the 4096 latents, *zero* were flagged by the scanner audit, with a mean dataset-prediction AUROC of 0.5037 across the dictionary — effectively at the random baseline of 0.5. This is consistent with the choice of tap point: the Stage 3→4 boundary lies sufficiently deep in the encoder that representations have abstracted away from acquisition-level signals such as scanner make, exposure, or institutional artefacts, and instead encode higher-level anatomical and pathological structure. No latent in the dictionary can predict the source dataset from its activation magnitude alone with AUROC above 0.8, so the audit removes no features and the full 4096-latent dictionary proceeds to the frequency stage.

4.1.2 Activation-Frequency Filter

Applying the $\geq 1\%$ frequency threshold on at least two of the four datasets reduces the candidate pool from 4096 to 863. Restricting to these L1-passing latents, Table 2 reports the mean per-dataset activation frequency.

Dataset	Mean activation frequency (%)
CheXpert	4.44
RSNA	4.53
SIIM-COVID-19	4.42
Shenzhen TB	3.72

Table 2: Mean per-dataset activation frequency over the 863 latents that passed the frequency audit. Computed as the average over latents of the per-dataset firing rate defined in Equation 10.

The CheXpert, RSNA, and SIIM-COVID-19 averages cluster within 0.1 percentage points of one another, indicating that the stratified-sampling and downsampling regime of Section 3.1 succeeds in preventing CheXpert’s volume advantage from dominating the SAE’s learned representations. Shenzhen TB lags by roughly 0.7 percentage points, suggesting that the smallest dataset in the corpus remains slightly under-represented in the harvested activation distribution even after sampling with replacement. Future work should explore alternative balancing strategies — per-dataset loss reweighting or longer over-sampled epochs — to close this residual gap.

4.1.3 Cross-Dataset Semantic Consistency

The centroid audit applied to the 863 frequency-passers reduces the pool to 279. For each candidate latent f , the per-dataset centroid $c_f^{(d)}$ of Equation 11 is computed as the mean encoder activation across the top-50 patches on which f fires in dataset d , and the minimum pairwise cosine similarity $\text{sim}(f)$ of Equation 12 is taken across all dataset pairs on which the latent fires. A latent is retained when $\text{sim}(f) \geq \tau_{\text{sem}} = 0.70$; the strict minimum, rather than a mean, ensures that a latent which is coherent on two datasets but diverges on a third is rejected as polysemantic across the corpus. Geometrically, this requires that the activation directions a latent responds to in different datasets are aligned in the 1024-dimensional encoder space, not merely that the latent fires often enough on each dataset. The drop from 863 to 279 shows that approximately two-thirds of frequency-passing latents fail this geometric consistency requirement: they fire often enough on multiple datasets, but on representationally distinct directions in each — the canonical signature of polysemantic features that have not been disentangled by the SAE.

4.1.4 Split-Feature Audit

The split-feature audit on the 279 semantically consistent candidates flags *no* pairs (f_a, f_b) at the largest schedule scale k_n . For each pair, the Jaccard similarity $J(\mathcal{T}_{f_a}^{(k_n)}, \mathcal{T}_{f_b}^{(k_n)})$ between the top-50 image sets of Equation 13 is computed; no pair exceeds the split threshold 0.5 of Equation 14, so no merges or ancestor lookups are required and the candidate count is preserved. This null result is itself informative: under a non-Matryoshka SAE, feature splitting is a documented failure mode in which the same concept is shared across two overlapping latents that each fire on a subset of the underlying images [6]. The absence of any flagged pair empirically supports the central design claim of the Matryoshka architecture — that the nested-prefix loss of Equation 9 regularises the dictionary against splitting [9] — and justifies its selection over flat alternatives such as BatchTopK for the universality task.

4.1.5 Prefix-Stability Audit

The final stage tests stability of the surviving 279 latents across the Matryoshka prefix schedule $k \in \{6, 12, 24, 48\}$, retaining only those latents whose top-image sets are consistent across nested SAE sizes. This filter removes a further 27 candidates, yielding 252 confirmed universal latents. These are the features submitted to the illness-stratified probing of Section 4.2, and they satisfy — by construction — every property of the universal-feature definition introduced in Section 1: domain invariance via the frequency audit, cross-population robustness via the scanner pre-filter, representational alignment via the centroid audit, and non-redundancy via the split and prefix-stability audits.

Table 3 summarises the surviving counts at each stage.

4.2 Illness Stratification

The probes of Section 3.4.6 were run on the 252 confirmed universal latents recovered by the universality pipeline of Section 3.4 on the Matryoshka SAE at the Stage 3→4 boundary. Of the four datasets, three contribute to the seven-way coarse probe: RSNA is excluded because its binary pneumonia label collapses six of the seven canonical illness groups into the same class, leaving the macro one-vs-rest AUROC undefined.

Stage	Threshold	Surviving
Initial dictionary	—	4096
Scanner-feature audit	$\text{AUROC} < 0.80$	4096
Activation-frequency filter	$\geq 1\%$ on ≥ 2 datasets	863
Cross-dataset semantic	$\text{sim}(f) \geq 0.70$ all pairs	279
Split-feature audit	$J < 0.5$ all pairs	279
Prefix-stability audit	Stable across $k \in \{6, 12, 24, 48\}$	252

Table 3: Funnel of confirmed universal latents through each stage of the pipeline of Section 3.4. The Matryoshka SAE produces 252 features that satisfy every audit, out of an initial dictionary of 4096.

4.2.1 Discriminability Distribution

Across the 252 confirmed latents, the cross-dataset mean AUROC $\overline{\text{AUROC}}_k$ is concentrated narrowly above the random baseline of 0.5, with most values in the range 0.54–0.58. The clinically informative signal lives in a small upper tail. Table 4 reports the count of latents exceeding three thresholds on at least one dataset and on all three multi-class datasets simultaneously.

Threshold	At least one dataset	All three datasets
$\overline{\text{AUROC}}_{k,d} > 0.60$	91 / 252 (36.1%)	3 / 252 (1.2%)
$\overline{\text{AUROC}}_{k,d} > 0.65$	40 / 252 (15.9%)	3 / 252 (1.2%)
$\overline{\text{AUROC}}_{k,d} > 0.70$	12 / 252 (4.8%)	2 / 252 (0.8%)

Table 4: Cross-dataset coarse-probe AUROC distribution over the 252 confirmed universal latents. Counts are per-dataset thresholds satisfied on at least one of CheXpert, SIIM-COVID-19, TB, and on all three.

The dataset on which discriminability is sharpest is SIIM-COVID-19: every top-tier latent achieves its highest single-dataset AUROC there. CheXpert yields the lowest peak score among the three, consistent with its multi-label structure — a single image can carry several pathologies, so even a perfectly disease-specific latent fires on multiple canonical groups. TB produces noisier estimates due to its small size (~ 700 images), visible as larger fold-to-fold variance $\sigma_{k,d}$.

4.2.2 Tier 1 — Backbone Latents

Six latents — 3512, 2640, 3321, 1969, 1084 and 3832 — fire on $\geq 99\%$ of tokens in every group on every dataset with nearly flat mean activation across groups, and achieve cross-dataset AUROC in $[0.54, 0.59]$. They occupy ranks one and two of $M_{g,k}$ for *all seven* canonical groups, which confirms the caveat introduced in Section 3.4.6: ranking by raw activation magnitude surfaces domain-level features — “this is a chest radiograph” — rather than clinically meaningful ones, and the AUROC-based ranking is the appropriate signal for clinical interpretation. After examining ablation heat maps it was evident that all of these features were focusing on the corners and edges of the images which is not of much value as it does not give any additional supporting evidence that these features have encoded some medical understanding. The six Tier 1 latents are therefore excluded

from the clinical interpretation that follows; the Tier 2 and Tier 3 analyses below operate only on latents whose AUROC-based ranking exceeds the domain-level baseline.

4.2.3 Tier 2 — Healthy-Lung Detectors

Seven latents fire most strongly and frequently on *no finding* images and are suppressed when any pathology is present (Table 5). Two stand out: latent 818 and latent 1384 are the only confirmed latents to achieve mean AUROC above 0.70 across all three multi-class datasets, with activation frequencies of $\geq 95\%$ on healthy images dropping to 52–88% on pathological ones.

Latent	CheXpert	SIIM-COVID	TB	Mean	Healthy freq
818	0.631	0.830	0.664	0.708	0.98
1384	0.622	0.843	0.634	0.700	0.95–0.96
884	0.596	0.740	0.529	0.622	0.97
2891	0.598	0.744	0.564	0.635	0.76
3757	0.618	0.680	0.564	0.621	0.92
2512	0.625	0.685	0.550	0.620	0.84
1632	0.567	0.760	0.525	0.617	0.86–0.90

Table 5: Tier 2 healthy-lung detectors. The healthy-frequency column reports activation frequency on the *no finding* group; sick-image frequencies fall to 0.17–0.74 on the same latents.

Latent 1384 attains the single highest dataset-level score in the entire confirmed set — AUROC = 0.843 on SIIM-COVID-19 — separating COVID-positive from COVID-negative radiographs from the magnitude of one scalar feature alone. Latent 2891 is structurally distinct: its activation magnitude is small (≤ 50), but its *frequency* ratio between healthy (0.76) and sick (0.22–0.38) is sharp enough that it functions as a near-binary indicator of healthy state rather than a graded detector.

4.2.4 Tier 3 — Pathology-Presence Detectors

Three latents show the opposite polarity: high activation on pathological groups, suppression on *no finding* (Table 6). Latent 2682 is the cleanest pathology-presence detector, firing on only 0.18–0.32 of healthy images but 0.47–0.70 of sick ones, and the inversion holds on all three datasets.

Latent	CheXpert	SIIM-COVID	Shenzhen TB	Mean	Peak group
2682	0.579	0.779	0.668	0.675	airspace opacity
3348	0.642	0.683	0.613	0.646	pleural abnormality
2619	0.614	0.708	0.522	0.615	cardiac/mediastinal

Table 6: Tier 3 pathology-presence detectors. “Peak group” is the canonical illness group with the largest z-score differential $\Delta_{g,k}$ between sick and healthy activations.

Latent 3348 attains both the highest CheXpert AUROC of any confirmed latent (0.642) and the largest single z-score differential observed in the entire analysis ($\Delta_{\text{pleural},3348} = 219.8$).

Latent 2619 peaks for *cardiac/mediastinal* ($\Delta = 185.3$) and is markedly weaker on TB (0.522), which is consistent with TB being a parenchymal rather than a cardiac disease.

4.2.5 Cross-Group Co-activation

Two co-activation patterns recur across the Tier 3 candidates identified by the z-score differential $\Delta_{g,k}$. First, latents 3348 and 1237 are top z-score standouts for both *pleural abnormality* and *atelectasis*; clinically, atelectasis (lung collapse) produces increased radiographic density that overlaps with pleural thickening or effusion, and the two conditions co-occur. Second, latents 2619 and 215 are top standouts for both *cardiac/mediastinal* and *vascular/edema*; left-sided cardiac failure simultaneously produces cardiomegaly and pulmonary vascular congestion. In both cases the SAE has recovered a clinically coherent co-occurrence pattern from the unsupervised reconstruction objective alone, without exposure to multi-label supervision.

4.2.6 Qualitative Inspection of the Strict Three

Figure 5 reports the patch-ablation maps of the three strict-criterion universal latents across the four chest X-ray datasets. Each panel stacks one row per dataset (CheXpert, RSNA, SIIM-COVID-19, Shenzhen TB): the left column shows the input radiograph with the top-activating patches outlined, and the right column shows the deletion-delta heatmap on the 7×7 token grid, with brighter regions indicating patches whose removal most strongly suppresses the latent’s activation.

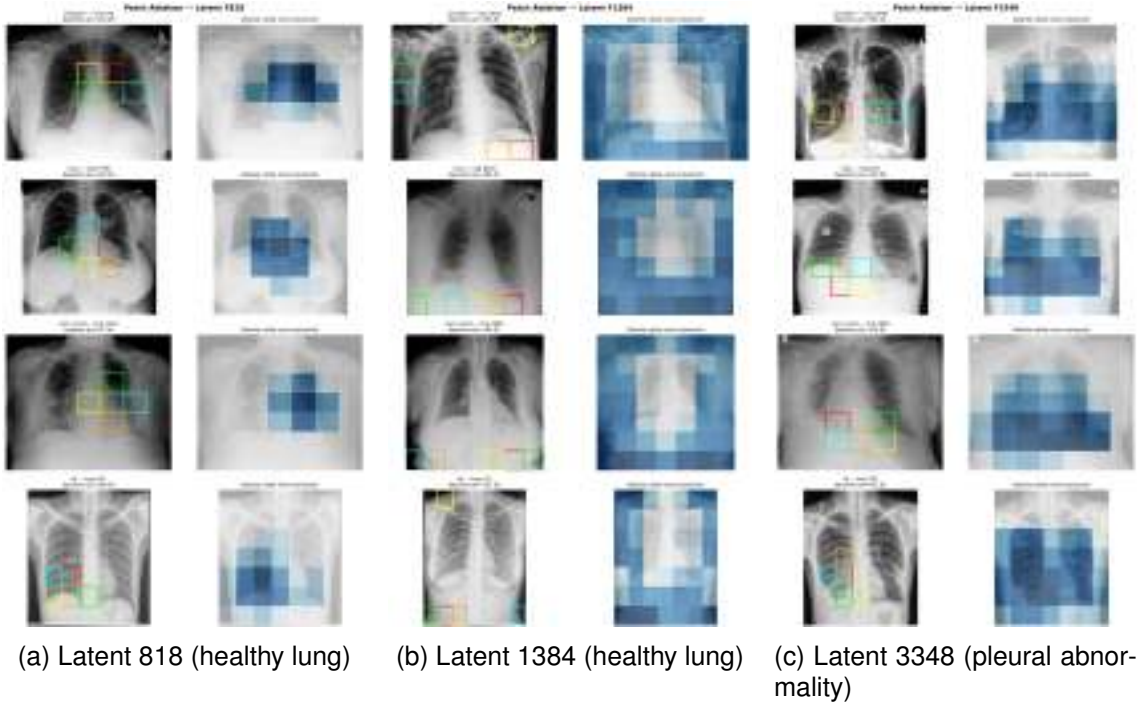


Figure 5: Patch-ablation maps of the three strict-criterion universal latents.

Two qualitative observations are worth noting. First, the deletion-delta maps are visually consistent across the four datasets within each latent: 818 and 1384 both concentrate their causal mass on the central thoracic region, and 3348 on the lower lung fields and pleural margins. This visual consistency is the same property that the centroid audit of

Section 3.4 measures numerically, and it provides direct evidence that the audit’s geometric criterion picks out latents whose dependence structure — not merely whose firing rate — is preserved across cohorts.

Second, latent 1384 displays a notable separation between its activation locus and its causal locus. Its top-activating patches sit at peripheral positions — lung apices, costophrenic angles — yet the deletion-delta heatmap concentrates on the central mediastinum and cardiac silhouette. The two readouts disagree, which is precisely the phenomenon flagged by Han et al. [21]: the causal effect of an SAE feature need not coincide with the inputs that maximally activate it. The Swin attention pattern at the Stage 3→4 boundary mixes information globally, so peripheral tokens can read out central anatomy. Latent 1384 is therefore better understood not as a peripheral-tissue detector but as a normal-mediastinal-configuration detector with a peripheral readout, which is consistent with its peak AUROC of 0.843 on the COVID-positive versus COVID-negative separation, where central opacification and mediastinal lymphadenopathy are dominant discriminators.

5 Conclusions

This thesis applied the Matryoshka sparse autoencoder of Bussmann et al. [9] to UMedPT activations harvested at the Stage 3→4 boundary on four chest X-ray datasets. A five-stage hierarchical pipeline (Section 3.4) reduced the 4096-latent dictionary to 252 confirmed universal candidates, which were then probed clinically through the illness-stratified linear probe of Section 3.4.6.

Three latents clear every stage under the strict criterion that the linear-probe AUROC exceeds 0.60 on *every* multi-class dataset. Latents 818 and 1384 are healthy-lung detectors firing on $\geq 95\%$ of *no finding* tokens, with mean AUROCs of 0.708 and 0.700; latent 3348 is a pleural-abnormality detector carrying the largest z-score differential in the confirmed set ($\Delta_{\text{pleural},3348} = 219.8$, mean AUROC 0.646). A larger pool of roughly twenty further latents clears the threshold on the multi-class mean but collapses to chance on at least one dataset — almost always Shenzhen TB — and therefore fails the strict bar even where the underlying concept is plausibly real.

The existence of these strict three is the central result of the work. UMedPT’s pretraining objective does not explicitly reward a shared clinical vocabulary, yet an unsupervised Matryoshka dictionary recovers a small population of features that behave as if one were present: healthy parenchyma, pleural abnormality, and a polarity contrast between the two. The Matryoshka architecture is material to this recovery, as both the split audit and the prefix-stability check depend on the nested-prefix structure.

6 Further Research

Several extensions follow. The pipeline is modality-agnostic and could be applied without modification to the MRI or histopathology subsets of the UMedPT corpus, the latter being a stronger test of universality precisely because its content is not legible to a non-specialist evaluator. A cross-modality study — asking whether a latent encoding pleural abnormality on radiographs also fires on the same condition under CT, or whether organ-level features transfer between modalities — would test the strong form of the universality claim directly. Lifting the 10,000-image quota of Section 3.1 and scaling the dictionary

width correspondingly would clarify whether the sparse population identified here is saturated or merely a lower bound.

A direct comparison against null baselines is also outstanding. The audit pipeline is not benchmarked against a randomly initialised Matryoshka SAE of identical architecture or a PCA projection of equal sparsity, so the absolute count of 252 confirmed universals cannot be compared with chance. The clinical-probe results of Section 4.2 provide indirect evidence that the population is non-trivial — random projections would not produce the pleural–atelectasis or cardiac–vascular co-activation patterns observed on Tier 3 latents, nor place the highest single-dataset AUROC on a COVID-positive versus COVID-negative separation — but a direct null comparison would settle the question.

As learned models are increasingly trusted in clinical settings, the ability to inspect and audit their internal computations stops being an academic concern. Sparse autoencoders are one of the few tools with a principled route from raw activations to human-legible concepts, and pushing that route into medical imaging is a useful test of how far it can be carried.

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